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Ecological inference and US elections

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Abstract

This thesis investigates the problem of ecological inference, which arises when we are interested in relationships between variables at the individual level but only have access to aggregate data.

Section 1 presents the problem of ecological inference, its relevance in various contexts and some of the solutions proposed. In particular, it presents the method proposed by McCartan and Kuriwaki (2026a), which is the one that will be used in this thesis.

Section presents a synthetic example to illustrate the method and its performance in a controlled setting, where the true individual-level relationships are known.

Section presents an application of the method to US elections, where we are interested in the relationship between individual-level characteristics and individual voting behavior, but we only have access to aggregate data.

Contents

1 Introduction

Ecological inference fallacy is the problem that arises when we are interested in relationships between variables at the individual level but we only observe data at the aggregate level.

This problem is relevant in all situations in which data are reported at the aggregate level, but we are interested in relationships at the individual level.

A famous example of that problem is the one presented by Robinson (1950) for the relationship between illiteracy and race, and illiteracy and migration in the United States. In that case, the author observed that areas with higher proportion of Black population, as well as areas with higher proportion of migrants, had lower illiteracy rates. However, when looking at the individual level, Black people and migrants were more likely to be illiterate than White and non-migrant people. The explanation of this apparent paradox is that Black and migrant people were more likely to live or move to areas with lower illiteracy rates, and therefore the aggregate data masked the individual-level relationship.

We can think also at economic examples. Consider, for instance, a retail store chain that wants to investigate consumers' habits of young people observing data about sells in different stores. The retail store chain might know quantities of goods sold in different stores and, at the same time, demographic data about consumers in different areas covered by different stores. The naive researcher, therefore, would link the percentage of young people and the quantities of different goods sold, suggesting that goods sold in stores with younger consumers are sold to young consumers. However, that is a clear example of ecological fallacy and it could lead to false findings. In fact, when we consider data about cities, it might well be that stores with greater share of young consumers are located in central areas near universities, where richer people also usually live.

Therefore, even if young people might be cheaper consumers, their habits would be masked by their neighbours'habits.

In any case, the most obvious case of ecological inference is in electoral behavior, where we are interested in the relationship between individual-level characteristics and individual voting behavior, but obviously (and luckily!) we are not able to observe individual-level voting behavior, and sometimes we are not able to observe individual-level characteristics either. In these cases, we are forced to rely on aggregate data, possibly choosing the most granular level of aggregation possible. Unfortunately, however, even choosing the most granular level of aggregation possible, individual-level relationships may be severely masked. That, of course, is higly relevant for political economists, who often times deal with the relationships between demographic characteristics, electoral results and public policies.

Kuriwaki and McCartan (2026) present a particularly convincing example of that problem, citing 1968 US presidential elections, where, in addition to the Republican and Democratic candidates Richard Nixon and Hubert Humphrey, there was also a relevant third-party candidate, George Wallace, who lead the segregationist American Independent Party. George Wallace had his strongest results in the Southern States, where the proportion of Black population was generally higher than in the rest of the nation. Therefore, a very naive analysis of the aggregate data would suggest that Black people were more likely to vote for George Wallace, whose platform was segregationist and overtly racist. Kuriwaki and McCartan (2026) go on to show that, even at more granular levels of aggregation, such as the county level, the relationship between proportion of Black population and proportion of votes for George Wallace would still be positive considering all counties in the nation; not only that, but even restricting the analysis to some specific States, such as North Carolina, we would still observe a positive relationship: the higher the proportion of Black population, the higher the proportion of votes for George Wallace.

The key here is that, while Black people were a quite homogenous group in terms of voting behavior (generally favoring the Democratic candidate and certainly not voting for George Wallace), White people were a much more heterogeneous group, and in particular, White people with segregationist views were also more likely to live in areas with higher proportion of Black population.

In fact, that same mechanism is at work in the retail store example: while young consumers might be quite homogenously cheap in their habits, the prevalent majority of older people has different habits and it drives overall difference.

Robinson (1950) came to the conclusion that ecological fallacy would constitute an inextricable problem and he concluded his paper telling that he was

> ... aware that this conclusion has serious consequences, and that its effect appears wholly negative because it throws serious doubt upon the validity of a number of important studies made in recent years. The purpose of this paper will have been accomplished, however, if it prevents the future computation of meaningless correlations and stimulates the study of similar problems with the use of meaningful correlations between the properties of individuals.

Econometrics in 1950, however, had considerably fewer instruments than it has now and we might think that some of these techniques could be successfully deployed in confronting ecological inference fallacy.

In particular, we could be tempted to argue that the issue could be solved by using covariates.

That, however, is not necessarily sufficient. In fact, when Kuriwaki and McCartan (2026) look at results in presidential elections in North Carolina, they are implicitly adding a covariate for the State and interacting it with the proportion of Black voters. Indeed, the estimate they present can be thought as the coefficient associated to the share of Black voters, given the covariate of the county being in North Carolina.

Analogously, we might include each possible interaction between variables and functional forms, for example imposing a non-linear relationship between Black voters' share and electoral results; that could result in a positive relationship between Black voters' share and segregationist votes for relatively small shares of Black voters that would later reverse when Black voters become majority.

Another method that we could possibly consider is the inclusion of fixed effects. However, fixed effects require repeated observations of the same unit, which is not the case for our examples.

All these examples highlight the common problem in adopting traditional econometric techniques to address ecological inference, which is the difference between the unit of observation and the quantity that we want to estimate.

In fact, we can think of each possible technique to better identify the relationship between variables at the level of unit of observation, but that would not solve the ecological fallacy of composition.

Indeed, even our naivest estimates might be correct in explaining relationships at the level of units of observations. It can be true that counties with more migrants have lower illiteracy rates even accounting for other variables or other effects, or that consumers in areas with younger people spend more in expensive goods, or that precincts with higher shares of Black voters favor segregationist candidates.

What these data can't tell us is whether migrants have lower or higher illiteracy rates, younger consumers spend more or less in expensive goods, or how Black people vote.

In order to answer these questions, we have to recover a notation that explicitly accounts for the aggregate nature of our units of observation and to impose a set of assumptions.

Only that will allow us to move from the agnostic conclusion of Robinson (1950) to ecological estimates.

In the next Section, I will introduce a general notation for ecological inference problems and I will briefly present some of the methods employed to address ecological inference and the set of assumptions they rely on.

In the following Section, I will present a more specific model for electoral analysis.

Both of these Sections will be heavily dependent on Kuriwaki and McCartan (2026), where the authors focus on theassessment of ecological inference methods, and McCartan and Kuriwaki (2026a), where the authors present the semiparametric method called seine. In any case, the notation that I will present is slightly different and more apt to describe the issues that we will face in this dissertation.

Once the model is presented, I will present a synthetic example to show how the method works and to assess its performance in a controlled environment.

Finally, I will present an application to real data about US presidential elections and I will conclude with some comments and conclusions.

2 Notation for ecological inference

We consider a population of N individuals, that we denote by $i \in [1, N]$.

Each individual i is characterized by a set of variables, which we can divide into a response variable Y_i and a categorical variable X_i.

The response variable can be either quantitative or categorical, and it will be denoted consequently. In case of a quantitative variable, we will denote it by a scalar $Y_i \in \mathbb{R}$. In case of a categorical variable, we will denote it by a vector $Y_i \in \{0, 1\}^p$, where p is the number of possible categories: that is, Y_i is a vector with all entries equal to 0 except for the entry corresponding to the value of the categorical variable for individual i, which is equal to 1.

That means that $Y_i = (1, 0, 0, ..., 0)$ if the response variable is the first category, $Y_i = (0, 1, 0, ..., 0)$ if the response variable is the second category, and so on...

The categorical variable X_i, instead, is consequently denoted by $X_i \in \{0, 1\}^d$, where d is the number of possible categories, and it encodes which is the category to which individual i belongs. We shall denote the categories with $k \in [1, d]$.

The total population is partitioned into G geographies, which we denote by $g \in [1, G]$: each individual belongs to one and only one geography, which we denote by G_i.

Geographies are what we previously referred to as units of observation, therefore we are never able to observe Y_i and X_i, but we only observe variables at geography level.

For each geography g, we have population $N_g = \sum_i \{1 | G_i = g\}$.

Moreover, we have, $\overline{Y}_g = \frac{1}{N_g} \sum_i \{Y_i | G_i = g\}$ and $\overline{X}_g = \frac{1}{N_g} \sum_i \{X_i | G_i = g\}$.

$\overline{Y_g}$ can be either a scalar, if the response variable is quantitative, or a vector of shares $\overline{Y_g} \in [0,1]^d$, if the response variable is categorical; in this second case, each entry of the vector would represent the share of the respective category and the sum of entries would be 1 as the categories are a partition of the population.

$\overline{X}_g$, instead, is a vector of shares $\overline{X}_g \in [0,1]^d$.

2.1 Global and local estimands

In general, we are interested in the average of Y_i conditional on the belonging to category $k \in [1, d]$.

We shall denote this quantity by $\beta_k = \frac{\sum_i \{Y_i | i \in I_k\}}{N_k}$, where I_k is the subset of people belonging to category k, while N_k is the cardinality of I_k.

As before, β_k is a scalar if the response variable is quantitative and a vector of shares if the response variable is categorical.

When we consider all different categories $k \in [1, d]$ we end up either with a vector $\beta \in \mathbb{R}^d$ or a matrix $\beta \in [0,1]^{p \times d}$.

We shall call β the global estimand.

However, as always, we do not observe individuals and therefore it is useful to recover quantities and definitions at geographical level.

The local estimand for category k and geography g is defined as $\beta_{gk} = \frac{\sum_i \{Y_i | i \in I_{gk}\}}{N_{gk}}$, where I_{gk} is the subset of people belonging to category k and geography g and N_{gk} is its cardinality.

Analogously, we can collect all β_{gk} pertaining to the same geography g but to different categories k within β_g of dimension d or $p \times d$.

These definitions allow us to derive two accounting identitities.

The first one links global estimand to local estimands: $\beta_k = \frac{\sum_{g=1}^{G} \beta_{gk} * N_{gk}}{N_k}$, i.e. the global estimand is the weighted average of local estimands, weighted by the population of each geography.

Therefore, this identity suggests a strategy that starts from local estimands, which can also be of interest on their own, to recover the global estimand.

The second accounting identity is at geographical level: $\overline{Y}_g = \beta_g' \overline{X}_g$.

One consequence of this second identity is that each observation contributes d or $p \times d$ parameters, making the local problem unidentifiable.

2.2 Different assumptions, different strategies

2.2.1 CCAR assumption

The easiest way to escape the identification issue is to assume that $\beta_g = \beta, \forall g$ and, consequently, adopt a simple linear regression, as the second accounting identity becomes $\overline{Y}_g = \beta' \overline{X}_g$.

This assumption, however, is particularly stringent and it can be easily falsified by data, for example by finding two geographies with same $\overline{X}_g$ but different $\overline{Y}_g$.

A slightly more relaxed assumption is allowing for variations in β_g, but with the constraint that variations in β_g are independent of $\overline{X}_g$: $E[\beta_g|\overline{X}_g] = \beta, \forall g$; or, equivalently, $E[(\beta_g - \beta)|\overline{X_g}] = 0, \forall g$.

That assumption, which cannot be falsified by observable data per se, would still allow us to use linear regression.

In fact, considering geographies g as units of observations and $\overline{Y}_g$ and $\overline{X}_g$, it would be equivalent to a traditional linear model of the following form: $\overline{Y}_g = \beta_g * \overline{X}_g = \beta * \overline{X}_g + \epsilon_g$, with $\epsilon_g \equiv (\beta_g - \beta) * \overline{X}_g$.

Thanks to the assumption that $E[(\beta_g - \beta)|\overline{X_g}] = 0$, we would have $E[\epsilon_g|\overline{X}_g] = E[(\beta_g - \beta) * \overline{X}_g|\overline{X}_g] = E[(\beta_g - \beta)|\overline{X_g}] * \overline{X_g} = 0$ and linear regression could be used.

As underlined by Kuriwaki and McCartan (2026), that is what Goodman (1953) described, even if with a more primitive notation, when discussing conditions that would allow to exploit linear regressions in ecological inference, overcoming Robinson (1950)'s worries.

This assumption is called Coarsening Completely At Random (CCAR), as it is equivalent to think that the way in which individuals are divided into geographies is independent of other characteristics of the geography: literate and illiterate White people, cheaper and more expensive older consumers, segregationist and non-segregationist White voters are randomly assigned to counties, retail stores and electoral precincts.

Equivalently to linear regression, CCAR would allow the following identification: $\beta = E[\overline{X}\overline{X}']^{-1}E[\overline{X}'\overline{Y}]$.

In any case, when performing ecological regression, it is hardly the case that this assumption holds true: in the examples presented in the Introduction, that would mean that White and non-migrants people shared the same average illiteracy rates (β_g) independently on the percentage of Black and migrant people in their county ($\overline{X}_g$), that older consumers shared spending habits (β_g) independently on the share of young people in their areas ($\overline{X}_g$) and that White shared the same voting behaviors (β_g) independently on the percentage of Black people in their precinct ($\overline{X}_g$).

Kuriwaki and McCartan (2026) put that in the following way:

> Is coarsening completely at random plausible in practice? Consider again the 1968 election example. In this case, CCAR means that the preference of White voters for Wallace is unrelated to the proportion of Black voters in a county or the population of the county. For instance, this means that all of the following groups support Wallace to the same extent in North Carolina in expectation: White voters in rural lowland counties with a higher proportion of Black voters,White voters in urban counties, and White voters in mountainous counties with few racial minorities. CCAR is implausible in this setting, and is incompatible with the social scientist's interest in how different individuals sort into different coarsened geographies.

2.2.2 Adding covariates: CAR assumption

What we have not discussed so far is the possibility of adding covariates to the model, which would allow to relax the CCAR assumption.

We shall now introduce a set of covariates Z_g at geography level. As geographies are our units of observation, covariates can either derive from aggregation of individual characteristics or be characteristics of the geography itself: this distinction is not relevant for our purposes, and we will later see examples of both types. A consequence on notation is that we denote covariates as Z_g without the overline, as they are not necessarily averages of individual characteristics.

Thanks to covariates, we are now able to relax the CCAR assumption and considers the following assumption instead: $E[\beta_g|\overline{X}_g, Z_g, N_g] = E[\beta_g|Z_g]$.

This assumption is called Coarsening At Random (CAR) and it is equivalent to think that the way in which individuals are divided into geographies is independent of other characteristics of the geography, conditional on covariates: literate and illiterate White people, cheaper and more expensive older consumers, segregationist and non-segregationist White voters are randomly assigned to counties, retail stores and electoral precincts, conditional on covariates.

As a consequence, geographies may have different β_g and, contrary to CCAR, $\overline{X}_g$ can affect β_g, but we assume that there exist covariates Z_g which are able to capture the effect of $\overline{X}_g$ on β_g.

Therefore, we are able to write β_g as a function of the covariates: $\beta_g = E[\beta_g|\overline{X}_g, Z_g, N_g] + \epsilon_g = E[\beta_g|Z_g] + \epsilon_g$, with $E[\epsilon_g|\overline{X}_g] = 0$.

Without loss of generality, we can express $E[\beta_g|Z_g]$ with a vector-valued function: $E[\beta_g|Z_g] = f(Z_g)$.

Then, thanks to the second accounting identity, we have a model that links $\overline{Y}_g$ to $\overline{X}_g$ and Z_g: $\overline{Y}_g = \beta_g'\overline{X}_g = f(Z_g)'\overline{X}_g + \epsilon_g'\overline{X}_g$.

Taking the conditional expectations on both sides, we have $E[\overline{Y}_G|\overline{X}_g, Z_g, N_g] = f(Z_g)'\overline{X_g}$.

That allows to make the problem identifiable: we do not have to estimate a different set of parameters β_g for each observation as it was suggested by the second accounting identity, but a single functional form $f(Z_g)$ that links geographical covariates and unobserved β_g.

The easiest way to perform estimation would be to impose a linear functional form for $f(Z_g)$ ($f(Z_g) = \gamma' Z_g$), making the whole model linear and therefore estimable with linear regression: $E[\overline{Y}_G|\overline{X}_g, Z_g, N_g] = f(Z_g)'\overline{X_g} = (\gamma' Z_g)'\overline{X_g} = Z_g'\gamma\overline{X}_g$.

As discussed in McCartan and Kuriwaki (2026a), this condition would be quite stringent, but it is not necessary.

In fact, McCartan and Kuriwaki (2026a)'s methodology is a form of semiparametric estimation which uses the linear relationship of the second accounting identity ($\overline{Y}_g = \beta_g' \overline{X}_g$) and then allows for nonparametric form of $f(Z_g)$, requiring only two additional assumptions.

These assumptions are defined as positivity and bounded N.

The first additional assumption requires sufficient variation in $\overline{X}_g$ after controlling for Z_g, and it is analogous to the overlap assumption in causal inference: for each value of Z_g, there is a non-zero probability of observing each value of $\overline{X}_g$.

In absence of this assumption, it would be impossible to disentangle the effect of $\overline{X}_g$ on β_g from the effect of Z_g on β_g, and therefore the problem would still be unidentified.

The second additional assumption requires that there is not an overwhelmingly large geography whose effect dominates the effect of other geographies.

In the next Section I will present the model for electoral competitions, and I will discuss how it satisfies these assumptions.

3 A model for electoral competitions

Electoral competitions can be thought of situations where each voter i chooses a candidate over p possible candidates so that the response variable Y_i is a p-dimensional vector $Y_i \in \{0,1\}^p$.

However, both in the synthetic example and in the application on US electoral data, I will consider Y_i as a binary choice: in the synthetic example that is a consequence of referendum rules, where only two choices (vote Yes or vote No) are possible, while in the empirical application that is a consequence of the American two-party system, where two parties dominate the political landscape and the other parties are not relevant in the analysis.

Therefore, we can select one of the two possible votes and denote Y_i by a simple binary indicator ($Y_i \in \{0,1\}$) which takes value 1 if the vote is cast in favor of that choice, and 0 otherwise.

Electoral precincts are the geographical units in which votes are aggregated, and they are the units of observation in our analysis. Therefore, we can denote by g a generic electoral precinct and by G the set of all electoral precincts.

As a consequence, the observed variable $\overline{Y}_g$ is a scalar $\overline{Y}_g \in [0,1]$ that represents the share of the reference choice in the electoral precinct g.

We then consider a partition of individuals into d mutually exclusive and exhaustive categories, which are the categories of the variable $X_i \in \{0,1\}^d$. One of such partitions could be the partition of voters into mutually exclusive and exhaustive income classes, or the partition of voters into mutually exclusive and exhaustive racial groups.

Finally, each electoral precinct g is characterized by a vector of covariates

$Z_g \in \mathbb{R}^k$, which can either be characteristics of the geography itself (e.g., their location in a particular region) or arise as aggregation of individual-level covariates (e.g., the education level of the voters in that precinct).

It is now necessary to introduce the assumptions that allow to identify the model and to estimate the parameters of interest.

The positivity assumption depends on the choice of covariates Z_g and on the absence of collinearity between them and $\overline{X}_g$ in order to make the model identifiable.

The bounded N assumption can be verified by verifying that there are no overwhelmingly large electoral precincts.

Finally, we can assume Coarsening At Random (CAR) assumption through an individual-level assumption, which, following McCartan and Kuriwaki (2026a), we can call CAR-IND.

CAR-IND states that the individual-level response variable Y_i is independent of the geography g conditional on the individual-level covariates X_i and the geography-level covariates Z_g. This can be stated as $E[Y_i|G_i = g, X_i = x_i, Z_{G_i} = z_g] = E[Y_i|X_i = x_i, Z_{G_i} = z_g]$.

From a practical point of view, this assumption implies that the covariates are able to capture all the effects of the geography on the individual-level response variable Y_i. Going back to the electoral example presented in previous Sections, this assumption implies that White voters in a precinct g with certain characteristics Z_g have the same probability of voting for segregationist candidate as White voters in a different precinct g' with the same characteristics $Z_{g'} = Z_g$.

Moreover, as $f(Z_g)$ is not necessarily injective, it is possible that two different electoral precincts g and g', even with different characteristics $Z_g \neq Z_{g'}$, have the same value of $f(Z_g) = f(Z_{g'})$, and therefore the same expected value of $\beta_g = \beta_{g'}$.

McCartan and Kuriwaki (2026a) prove that CAR-IND implies CAR.

Therefore, in order to perform semiparametric estimation of the model, we need to check for bounded N and later select covariates Z_g which can satisfy the testable positivity assumption and the untestable CAR-IND assumption.

4 A synthetic example

In this section I will show a synthetic example that helps explaining why the problem is relevant and shows the difference perfomance between naive methods and the semiparametric estimation proposed by McCartan and Kuriwaki (2026a) and embedded in R package `seine` (McCartan and Kuriwaki (2026b)).

In the synthetic example we are the ones who generate the data and therefore we know the exact model of the reality. We show that, if we generate the data and then we try to estimate with different methods we obtain different results and the naive methods do not work well.

4.1 Italian constitutional referendum: a motivating example

In particular, we can exploit the constitutional referendum in Italy to show a possible way in which the naive estimators could not work and therefore we can generate a dataset that encompasses the problem.

The problem is the following: we want to estimate how did voters with different levels of income vote.

If we look at data at regional level, we would be tempted to say that richest people voted more in favour of the government proposal. In fact, the only three regions where the votes in favour were more than the votes in contrast (Lombardy, Veneto and Friuli-Venezia Giulia) are among the richest regions in Italy. On the contrary, the regions were No got more than 60% are among the poorest regions of Italy (Basilicata, Campania, Sicily).

However, once we look at data at municipal level, we find that, within the regions, the relationship is nearly inverse: No votes were more frequent in urban areas and other municipalities where income is relatively higher.

That divergence makes us think that if we were able to observe results at individual level, we could find even different relationships between income and voting behavior.

Therefore, we can now generate a synthetic dataset that presents a structure that resembles some stylized facts that we observe in the constitutional referendum results and other notions that we have about electoral behaviors in Italy.

4.2 The synthetic experiment

The synthetic experiment is constructed as follows.

We assume that election takes place in a Country which has three different regions: North, Centre and South.

Each region is divided in 100 precincts, so that in the end we observe 300 precincts: 100 precincts in the North region, 100 in the Centre region, 100 in the South region.

Each precinct has the same voting population N_g, which is then normalised to 1.

For each electoral precinct, we observe the percentage of high-income and low-income voters ($\overline{X}_g$) and the electoral results (Y_g).

Within each income category, there is high variability in their electoral behaviour across different precincts.

In particular, we impose that high-income people in the North and South precincts have a high propensity of voting in favour of the reform, while in the Centre this percentage is much lower. As for low-income people, we impose that in North they have high propensity of voting in favour, while in the Centre and in the South this propensity is much lower.

At the same time, we impose that the percentage of high-income voters is highest in the North and lowest in the South and intermediate in the Centre.

4.2.1 Data generating process

We generate the data as follows.

For each region $r \in \{\text{North}, \text{Centre}, \text{South}\}$ and for each of the 100 precincts in that region, we draw the share of high-income voters from a truncated normal distribution,

$$\overline{X}_{gr}^{H} \sim \text{TruncNormal}(a = 0, b = 1, \mu_r, \sigma = 0.2),$$

with means $\mu_{\text{North}} = 0.7$, $\mu_{\text{Centre}} = 0.5$, and $\mu_{\text{South}} = 0.3$. The low-income share is then set to $\overline{X}_{gr}^{L} = 1 - \overline{X}_{gr}^{H}$. Hence, on average, Northern precincts have higher shares of high-income voters, Southern precincts have lower shares, and Central precincts lie in between. The use of Truncated Normal distribution allows to have quite high variability within geography ($\sigma = 0.2$), but without having negative shares or shares above 1.

Subsequently, I assign to each precinct a propensity to vote in favour of the reform within each income group and I encode it in a vector $\beta_g = \begin{pmatrix} \beta_g^H \\ \beta_g^L \end{pmatrix}$.

β_g^H and β_g^L are extracted again from truncated normal distributions with different means depending on the region.

$$\beta_{gr}^{H} \sim \text{TruncNormal}(a = 0, b = 1, \mu = \eta_{gr}^{H}, \sigma = 0.05),$$
$$\beta_{gr}^{L} \sim \text{TruncNormal}(a = 0, b = 1, \mu = \eta_{gr}^{L}, \sigma = 0.05).$$

with means η^H_{gr} and η^L_{gr} defined as follows:

$$\eta^H_{\text{North}} = 0.6, \quad \eta^L_{\text{North}} = 0.7,$$
$$\eta^H_{\text{Centre}} = 0.3, \quad \eta^L_{\text{Centre}} = 0.3,$$
$$\eta^H_{\text{South}} = 0.6, \quad \eta^L_{\text{South}} = 0.2.$$

The key feature of this design is that the income-specific propensities are heterogeneous within each group and also vary across regions. High-income voters in the North and South have a strong tendency to support the reform, while in the Centre the same group is much less favourable. Low-income voters have a relatively high propensity to support the reform in the North, but not in the Centre or South.

Finally, the observed precinct-level outcome is generated as the weighted average of these two group-specific propensities:

$$Y_g = \overline{X}^H_g \beta^H_g + \overline{X}^L_g \beta^L_g.$$

In the end, we have a dataset that for each electoral precinct g contains the following information: the share of high-income voters $\overline{X}^H_g$, the share of low-income voters $\overline{X}^L_g$, the observed outcome Y_g, the location of the precinct (North, Centre or South), the expected propensity to vote in favour of the reform for high-income voters η^H_g and for low-income voters η^L_g (equal across geographies of the same region), and the actual propensities β^H_g and β^L_g (specific of each geography).

4.2.2 From DGP to the model

The generated dataset strictly satisfies the assumptions stated in previous Sections.

The response variable Y_i is a binary variable $Y_i \in \{0,1\}$ and it is observed only at geography level as a share $\overline{Y}_g \in [0,1]$.

Population is partitioned across geographies and within each geography, it is partitioned into two groups (high-income and low-income voters), so that for each geography g we observe $\overline{X}_g = \begin{pmatrix} \overline{X}_g^H \\ \overline{X}_g^L \end{pmatrix}$ with $\overline{X}_g^L = 1 - \overline{X}_g^H$.

Moreover, we can interpret the region of each precinct as a covariate $Z_g = \{0,1\}^3$, which in this case does not arise from individual characteristics.

$\overline{Y}_g$, $\overline{X}_g$ and Z_g are observed, while β_g (and their corresponding η_g) are unobserved and are the goals of the semiparametric estimation.

Bounded N is satisfied as N_g is equal across different geographies.

Positivity is satisfied as we have allowed for variability in the shares of high-income and low-income voters across geographies within each region.

Finally, CAR-IND is almost completely satisfied, thanks to the fact that β_g are generated from a distribution with low variance ($\sigma = 0.05$) that depends on the covariate Z_g.

In fact, a complete adherence to CAR-IND would require β_g to be equal across geographies of the same region, but I have allowed for some variability in order to make the example more realistic and more responding to the expected ability of the covariates to explain the variation in the propensitities.

In our case, $E[Y_i|G_i = g, X_i = x_i, Z_{G_i} = z_g] = \beta_g$ is just slightly different from $E[Y_i|X_i = x_i, Z_{G_i} = z_g] = \eta_{gr}$.

4.3 Comparison of results

4.3.1 OLS estimates

As previously mentioned, in our dataset we know the true values of β_g for each geography g, therefore we are able to obtain the true values of the estimands of interest β^H and β^L by aggregating the true values of β_g as stated in the first accounting identity in Section 2: $\beta^H = \frac{\sum_{g=1}^{G} \beta_g^H * N_g^H}{\sum_g N_g^H}$ and $\beta^L = \frac{\sum_{g=1}^{G} \beta_g * N_g^L}{\sum_g N_g^L}$.

In our case, we have that $\beta_H = 0.503$ and $\beta_L = 0.343$.

The naive OLS specification, implicitly assuming CCAR, would rely on the following regression: $\overline{Y_g} = \beta_H * \overline{X}_g^H + \beta_L * \overline{X}_g^L + \epsilon_g$. As $\overline{X}_g^H$ and $\overline{X_g^L}$ are perfectly collinear($\overline{X}_g^H + \overline{X}_g^L = 1$), we use $\overline{X}_g^L$ as reference category and we estimate the following regression:

$$
\begin{aligned}
\overline{Y_g} &= \beta_H * \overline{X}_g^H + \beta_L * \overline{X}_g^L + \epsilon_g \\
&= \beta_H * \overline{X}_g^H + \beta_L * (1 - \overline{X}_g^H) + \epsilon_g \\
&= \beta_L + (\beta_H - \beta_L) * \overline{X}_g^H + \epsilon_g.
\end{aligned}
$$

Results are presented in Tab. 1 and show that β_L is estimated to be 0.213, while β_H is estimated to be 0.213+0.419 = 0.632.

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	0.213	0.018	12.167	0.000
high_income	0.419	0.032	13.078	0.000

Tab. 1: Naive OLS regression summary

That means that the naive OLS regression overestimates the propensity of high-income voters to vote in favour of the reform and underestimates the propensity of low-income voters to vote in favour of the reform, showing a more polarized distribution of votes conditional on income.

That can be explained by the fact that the naive OLS attributes higher percentages of votes in favour in Northern precincts only to their higher share of high-income voters, while in reality the higher percentages of votes in favour are also driven by the fact that low-income voters in Northern precincts have a higher propensity to vote in favour of the reform than low-income voters in other regions.

4.3.2 `seine` estimates

When we move to semiparametric estimation performed by `seine`, we obtain the results presented in Tab. 2, which are much closer to the true values.

predictor	outcome	estimate	std.error	conf.low	conf.high
high_income	outcome	0.504	0.018	0.469	0.539
low_income	outcome	0.341	0.011	0.319	0.363

Tab. 2: Semiparametric ecological inference estimates

5 An application to Texas electoral data

After having showed the differential performance of `seine` and naive methods in a simple synthetic example of electoral competition, I moved to an application to real data.

In particular, I applied the semiparametric estimation to the 2020 US presidential election in Texas, using the electoral precincts as the geographical units.

5.1 Construction of the dataset

The construction of the dataset is explained in detail in Appendix A (Section 8).

In particular, I exploited three sources:

- ALARM dataset by T. Kenny and McCartan (2021), which combines electoral results at the VTD level previously collected by Voting and Election Science Team (2020) with demographic data of 2020 US census collected by U.S. Census Bureau (2020)
- ACS surveys, which provide socio-economic characteristics of the population and can be called through the `tidycensus` package in R (Walker and Herman, 2026)
- U.S. Census Bureau (2021)'s files, which provide the mapping of census blocks to CBGs and VTDs, which is necessary to aggregate the socio-economic characteristics collected at the CBG level in ACS surveys to the VTD level.

5.2 Choice of variables and validity of the assumptions

5.3 Results of the estimation

6 Other Comments

Other comments adesso facciamo una prova

7 Conclusions

Conclusions

References

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8 Appendix A: Construction of the dataset

The dataset to perform the analysis should include, at the most granular level possible, the share of votes received by the candidate ($\overline{Y}_g$), a partition of the voting population in categories of interest (e.g, ethnic categories, which represent $\overline{X}_g$) and covariates at that same geographical level (Z_g).

The most granular level for electoral results in the United States is the electoral precinct, whose dimension is typically in the order of hundreds of a couple of thousands of voters.

Census data, instead, are collected at different geographical levels, with geographical levels varying with the variables of interest.

The most granular level for census data is the census block, which is a geographical unit that is typically bounded by geographical features either natural (e.g., rivers) or artificial (e.g., streets). Given their geographical definition, census blocks are quite different in population size, with a considerable amount of blocks being uninhabited, and the most populated ones (typically in urban areas) having a population of hundreds of people.

The only variables collected at the census block level are the total population, the voting age population, ethnic composition and the number of housing units. Moreover, these variables are collected at the census block level only in the decennial census, which is conducted every ten years.

Census blocks are then grouped into larger geographical units, which are used to collect other variables of interest. In particular, the census block groups (CBGs) are the smallest geographical unit for which the American Community Survey (ACS) collects data on a yearly basis. CBGs are typically composed of a few hundred to a few thousand people, and they are designed to be relatively homogeneous in terms of population characteristics.

Variables collected at the CBG level in ACS include several socio-economic characteristics, such as income, educational attainment, employment status, as well as language spoken at home, commuting patterns, and housing characteristics. The ACS data are collected through sample surveys, so that data at the CBG level are subject to sampling variability.

Finally, census block groups are grouped into larger geographical units, which are called census tracts.

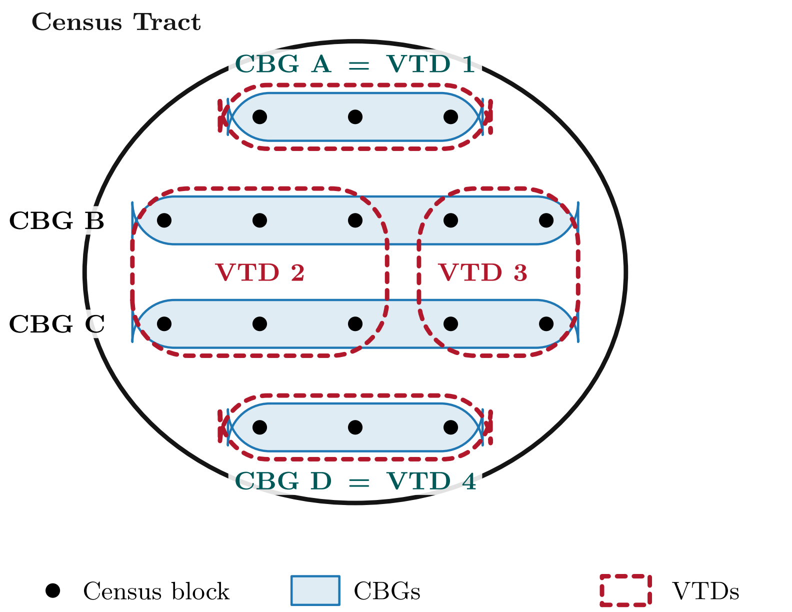
The electoral precincts, instead, are geographical units that are defined by the electoral authorities for the purpose of organizing elections. Precincts are typically composed of a few hundred to a few thousand voters, and they are sometimes denoted as voting districts (VTDs).

VTDs, however, are not units of the census, and they are not completely aligned with CBGs. VTDs are always constituted by a set of census blocks, and they are contained within a single census tract, but they can cross the boundaries of CBGs. This means that a VTD can contain parts of multiple CBGs, and a CBG can be split across multiple VTDs; viceversa is also true.

Fig. 1 shows a schematic representation of a possible relationship between census blocks, CBGs, census tracts and VTDs.

Some of the variables of possible interest for the analysis are collected only at the CBG level in ACS surveys. Therefore, a procedure to aggregate the data at the VTD level was necessary.

ALARM dataset by T. Kenny and McCartan (2021) combine electoral results at the VTD level previously collected by Voting and Election Science Team (2020) with demographic data of 2020 US census collected by U.S. Census



Census blocks are the atomic units. CBGs and VTDs are two independent partitions of the same census tract, using census blocks as atomic units. VTD 1 and VTD 4 coincide exactly with CBG A and CBG D, while VTD 2 and VTD 3 cut across CBG B and CBG C. Both partitions are entirely contained within the census tract.

Fig. 1: Relationship between electoral precincts (VTDs), census blocks, census block groups (CBGs), and census tracts.

Bureau (2020).

That provides a dataset that includes the share of votes received by the candidates at the VTD level, as well as the total population and the voting age population at the census block level with the corresponding ethnic composition.

This dataset, while being very useful, does not include the socio-economic characteristics, which are of fundamental importance for the analysis, and which are collected at the CBG level in ACS surveys. ACS data are easily accessible through the `tidycensus` package in R (Walker and Herman, 2026).

The following passage involved exploiting U.S. Census Bureau (2021)'s files, which provide the mapping of census blocks to CBGs and VTDs, to aggregate the socio-economic characteristics collected at the CBG level in ACS surveys to the VTD level.

As in Fig. 1, we can think of three scenarios: VTDs that perfectly align with one CBG, VTDs that aling with exactly two or more CBGs, and VTDs that cross the boundaries of CBGs.

For the first scenario, the aggregation of socio-economic characteristics from CBGs to VTDs is straightforward: the characteristics of the VTD are simply the same as those of the corresponding CBG.

The second scenario is also straightforward: the socio-economic characteristics of the VTD are simply the either the sum (in case of counts, e.g, population or the number of people with a college degree) or the average (in case of other quantities, e.g. median income) of the socio-economic characteristics of the CBGs that compose it.

The third scenario is a bit more complicated and required a procedure to apportion the socio-economic characteristics of the CBGs that are crossed by the VTD boundaries.

The procedure rested on the assumption that that differences in the apportionment of census blocks in the construction of the VTDs and CBGs are independent of the socio-economic characteristics of the population.

That assumption is partly disputable, because the electoral authorities in the United States have been known to engage in gerrymandering, which is the practice of drawing electoral district boundaries in a way that favors a particular political party or group. However, gerrymandering is typically done at the level of congressional districts, without particular interventions at the level of VTDs.

Moreover, census block groups are designed to be relatively homogeneous in terms of population characteristics, so that the differences in apportionment of census blocks between VTDs and CBGs are likely to be small.

Therefore, I have assumed that the average socio-economic characteristics of the population in a CBG are the same as those of the population in the census blocks that compose it. Subsequently, I have constructed fictitious populations of the census blocks, assigning to each census block the average socio-economic characteristics of the population in the corresponding CBG.

I will give two examples of the procedure, one for a count variable and one for a non-count variable: if a CBG with population of 1000 people has 200 people with a college degree, I have assumed that each census block in that CBG has 20% of its population with a college degree, so that a census block with 100 people has 20 people with a college degree; on the other side, if a CBG has a income of $50,000, I have assumed that each census block in that CBG has a median income of $50,000.

Finally, I have aggregated the fictitious populations of the census blocks to the VTD level, obtaining an estimate of the socio-economic characteristics of the population in each VTD. Obviously, these estimates are subject to some error, partly arising from the sampling errors in the original ACS data, and partly from the apportionment procedure.

However, this procedure is easily implementable and can be applied to any variable of interest that is collected at the CBG level.

9 Appendix B: A non-linear synthetic example

I might suppose that Z_g is a continuous variable, for example the share of immigrants within a precinct. Therefore, there exist no regional distinction, precincts differ by their shares of immigrants. However, we have a relationship, even if loose, between the share of immigrants and $\overline{X}_g$.

We should proceed as follows:

extract $\overline{X}_g$ for each geography from a random distribution

(e.g. $\overline{X}_g \sim \text{TruncNormal}(a = 0, b = 1, \mu = 0.5, \sigma = 0.2)$)

that means that precincts differ for their percentage of high-income and low-income voters.

$Z_g \sim \text{TruncNormal}(a = 0, b = 1, \mu = 1 - \overline{X}_g, \sigma = 0.2)$

that means that there is a loose association between share of high-income voters and share of immigrants (interpretation: immigrants are more likely to live in low-income areas).

Then, we might assume that

$$\beta^H_{gr} \sim \text{TruncNormal}(a = 0, b = 1, \mu = \eta^H_{gr}, \sigma = 0.05),$$
$$\beta^L_{gr} \sim \text{TruncNormal}(a = 0, b = 1, \mu = \eta^L_{gr}, \sigma = 0.05).$$

$$\eta^H_{gr} = 10^{Z_g - 1}, \quad \eta^L_{gr} = e^{Z_g - 1},$$

interpretation: both high-income and low-income voters tend to favor the reform more as the share of immigrants in their precinct increases, but the effect is stronger for low-income voters and the relationship is non-linear

we should observe that a linear estimator is not apt, while `seine` is able to capture the non-linear relationship and provide good estimates of β_H and β_L